

Chapter 2: Non-Parametric Procedures

Modelling Survival Data in Medical Research

Collett (2023), 4th ed.

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Part 1

1. What Are We Doing in Chapter 2?
2. Estimating the Survivor Function $S(t)$
3. Q&A

Starting Point: We Have Survival Data. Now What?

Imagine you are a researcher. You have just collected data on 18 women who used an IUD. Some discontinued early, some were still using it when the study ended.

You have a list of numbers. What questions do you want to answer?

Questions about one group

- ▶ What fraction of women are still using the IUD after 1 year?
- ▶ What is the *median* time before discontinuation?
- ▶ How does the risk of discontinuation change over time?

Questions about two groups

- ▶ Do breast cancer patients with positive HPA staining survive shorter than those with negative staining?
- ▶ Is the difference real, or just random chance?

Chapter 2 gives us the tools to answer all of these questions **without** assuming any particular probability distribution.

What Does “Non-Parametric” Mean?

Parametric method

Assumes the data come from a specific distribution (e.g. exponential, Weibull).

Works well if the assumption is correct.

Fails badly if it is wrong.

Non-parametric method

Makes **no assumption** about the shape of the distribution.

Works in many situations.

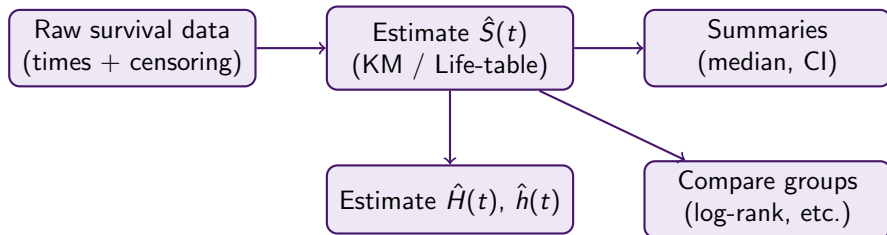
Does not rely on untestable assumptions.

Everyday analogy

Parametric: “Heights follow a normal distribution.”

Non-parametric: rank the heights, take the middle — no assumption needed.

The Big Picture of Chapter 2



Everything in Chapter 2 starts with estimating $\hat{S}(t)$. Get that right and the rest follows.

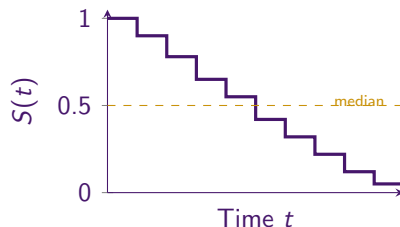
Recall: What Is the Survivor Function?

Definition from Chapter 1

$S(t) = P(T \geq t)$ = probability that an individual survives *beyond* time t

Key properties (recap):

- ▶ $S(0) = 1$: everyone is alive at the start
- ▶ $S(t)$ decreases over time
- ▶ $S(t) \rightarrow 0$ eventually
- ▶ The shape tells us about the risk



The Simplest Case: No Censoring

If **everyone** in the study is followed until the event occurs (no one is lost), then estimating $S(t)$ is straightforward.

Empirical Survivor Function

$$\hat{S}(t) = \frac{\text{Number of individuals with survival time } \geq t}{\text{Total number of individuals in the study}}$$

Example: 5 patients: survival times 2, 4, 6, 8, 10 months

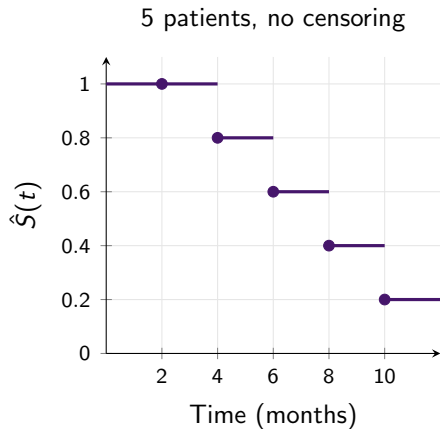
t	No. with $T \geq t$	$\hat{S}(t)$
$t < 2$	5	1.00
2	5	1.00
4	4	0.80
6	3	0.60
8	2	0.40
10	1	0.20

The Empirical Survivor Function is a Step Function

- ▶ $\hat{S}(t)$ stays **flat** between deaths
- ▶ It **drops** immediately after each death time
- ▶ Each drop equals $1/n$ (one person out of n)
- ▶ The result looks like a staircase going down

Only works without censoring

If patients are lost to follow-up or the study ends before their event, use KM instead.

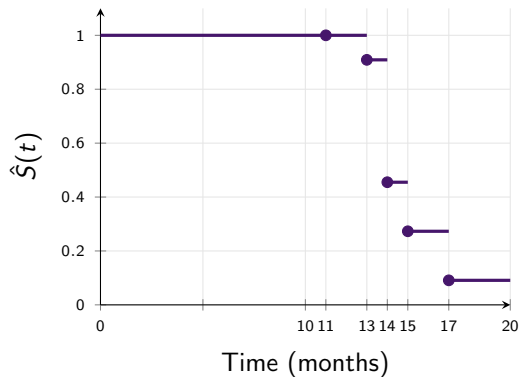


Example 2.1 — Pulmonary Metastasis

Burdette & Gehan (1970): 11 male patients with osteosarcoma

Survival times (months): 11 13 13 13 13 13 14 14 15 15 17

t	No. $\geq t$	$\hat{S}(t)$
< 11	11	1.000
11	11	1.000
13	10	0.909
14	5	0.455
15	3	0.273
17	1	0.091



Why Censoring Breaks the Simple Method

The problem with censored observations

Suppose patient A is censored at 50 weeks (study ended, still event-free).
When computing $\hat{S}(60)$, should we count patient A in the denominator?

Wrong approach: ignore censored

Drop all censored patients.
Now the denominator is too small.

⇒ We **overestimate** the probability of death

⇒ We **underestimate** $S(t)$

The estimate is **biased**.

Correct approach

Patient A tells us she survived **at least** 50 weeks.

She should be counted in the denominator for $t \leq 50$, but not beyond.

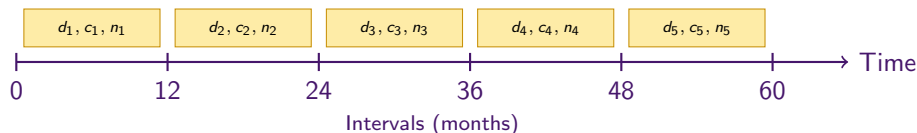
She *leaves the risk set* at $t = 50$.

This is what the KM estimate does.

The Life-Table Estimate — Main Idea

Strategy: divide time into intervals

Instead of tracking every individual death time, we group data into m time intervals. This was the standard method before computers.



In each interval, record:

- ▶ d_j = number of **deaths**
- ▶ c_j = number of **censored** observations
- ▶ n_j = number **at risk** at start

The Actuarial Assumption

Problem: what to do with censored observations within an interval?

We do not know *when* in the interval a censored observation occurred. So we assume they were censored **uniformly** across the interval — on average, at the midpoint.

Adjusted number at risk

$$n'_j = n_j - \frac{c_j}{2}$$

We subtract *half* the censored observations, because on average they were at risk for half the interval.

Example

$n_1 = 48, c_1 = 4 \Rightarrow n'_1 = 46.$ Survival prob: $(46 - 16)/46 = 0.6522.$

Life-Table Formula

Survival probability through interval j

$$\frac{n'_j - d_j}{n'_j}$$

This is 1 minus the estimated probability of dying in interval j .

Life-Table Estimate — running product

$$S^*(t) = \prod_{i=1}^{j-1} \frac{n'_i - d_i}{n'_i}, \quad t'_{j-1} \leq t < t'_j$$

Multiply together the survival probabilities from the *first* interval up to interval $j - 1$.

Think of it like this: to survive to month 24, you first have to survive month 0–12, *then* survive month 12–24. The probability of doing both is their product.

Example 2.2 — Multiple Myeloma (Life-Table)

48 multiple myeloma patients — grouped into 6 time intervals

Int.	Period	d_j	c_j	n_j	n'_j	$(n'_j - d_j)/n'_j$	$S^*(t)$
1	0–12	16	4	48	46.0	0.6522	1.0000
2	12–24	10	4	28	26.0	0.6154	0.6522
3	24–36	1	0	14	14.0	0.9286	0.4013
4	36–48	3	1	13	12.5	0.7600	0.3727
5	48–60	2	2	9	8.0	0.7500	0.2832
6	60–96	4	1	5	4.5	0.1111	0.2124

Life-Table — Step-by-Step Calculation

How to read and compute Table 2.1

- ① $S^*(t) = 1$ at the start (everyone alive)
- ② Interval 1: $S^*(12) = 1.000 \times 0.6522 = 0.6522$
- ③ Interval 2: $S^*(24) = 0.6522 \times 0.6154 = 0.4013$
- ④ Interval 3: $S^*(36) = 0.4013 \times 0.9286 = 0.3727$ (*continue multiplying ...*)

Interpretation

$S^*(24) = 0.4013$ means approximately **40%** of multiple myeloma patients are estimated to survive beyond 24 months from diagnosis.

Limitation

The estimate depends on the interval choice. When exact times are known, use **Kaplan-Meier**.

The Kaplan-Meier Estimate — The Gold Standard

Main idea

Instead of grouping into intervals, create an interval **for each individual death time**. This avoids any information loss from grouping.

Set-up:

- ▶ n individuals, r ordered death times:
 $t_{(1)} < \dots < t_{(r)}$
- ▶ At $t_{(j)}$: n_j at risk, d_j die
- ▶ Censored individuals leave the risk set without dying

The key step at each death time:

$$\text{Survival prob} = \frac{n_j - d_j}{n_j}$$

“Of those still alive just before time $t_{(j)}$, what fraction survived past $t_{(j)}$?”

Also called the **product-limit estimate** — it is the limiting case of the life-table as interval widths shrink to zero.

Kaplan-Meier — The Formula

Kaplan-Meier estimate

$$\hat{S}(t) = \prod_{j: t_{(j)} \leq t} \frac{n_j - d_j}{n_j}$$

Take the product of all survival factors at death times *up to and including* t .

What happens between death times?

$\hat{S}(t)$ is **constant** between adjacent death times.

Censored times do not cause a drop — they only reduce the risk set.

The KM estimate is the **product-limit estimate**: a product of small survival probabilities.

What happens at the end of the follow-up period?

Two scenarios depending on what the *largest* observed time is:

Largest observation is a **death**

$n_r = d_r$, so $(n_r - d_r)/n_r = 0$.

$\hat{S}(t) = 0$ from $t_{(r)}$ onwards.

The curve *reaches zero* at the last death time.

Largest observation is **censored**

No death at the end — the curve cannot drop.

$\hat{S}(t)$ is **undefined** beyond the last censored time.

The curve stops *without reaching zero*.

IUD data

Largest observation: $t = 107$ weeks, censored. So $\hat{S}(t)$ is undefined beyond 107 weeks.

Kaplan-Meier — Worked Example Step by Step

Tiny example: 6 patients

Death times: 3, 6, 10. Censored times: 5*, 8*, 12*.

Time	Event?	n_j	d_j	$(n_j - d_j)/n_j$	$\hat{S}(t)$
3	Death	6	1	$5/6 = 0.833$	$1.000 \times 0.833 = 0.833$
5	Censored	—	—	—	stays 0.833
6	Death	4	1	$3/4 = 0.750$	$0.833 \times 0.750 = 0.625$
8	Censored	—	—	—	stays 0.625
10	Death	2	1	$1/2 = 0.500$	$0.625 \times 0.500 = 0.313$
12	Censored	—	—	—	undefined beyond 12

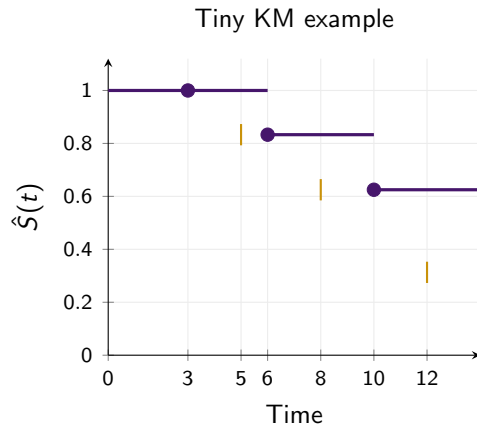
At $t = 6$, $n_j = 4$. Patient censored at $t = 5$ already *left the risk set* before this death time.

Kaplan-Meier — What the Step Function Looks Like

- ▶ The curve starts at $\hat{S}(0) = 1$
- ▶ It **drops** at each death time
- ▶ Censored observations are shown with a **tick mark** on the curve
- ▶ Between death times, the curve is **flat**
- ▶ The size of each drop depends on how many died *relative to* those at risk

Reading the curve

To find $\hat{S}(t)$: read the *height* of the step function at time t .



Example 2.3 — IUD Data: Kaplan-Meier Table

18 women, time to IUD discontinuation (weeks); * = censored

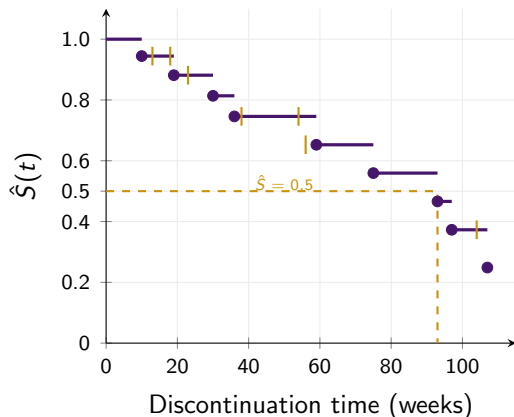
Death time	n_j	d_j	$(n_j - d_j)/n_j$	$\hat{S}(t)$
10	18	1	0.9444	0.9444
19	15	1	0.9333	0.8815
30	13	1	0.9231	0.8137
36	12	1	0.9167	0.7459
59	8	1	0.8750	0.6526
75	7	1	0.8571	0.5594
93	6	1	0.8333	0.4662
97	5	1	0.8000	0.3729
107	3	1	0.6667	0.2486

n_j drops 18→15 between $t = 10$ and $t = 19$: three women censored at 13, 18, 23 weeks.

Example 2.3 — IUD Data: KM Plot

Reading the plot:

- ▶ At 36 weeks: $\hat{S}(36) = 0.746$
 $\approx 75\%$ still using the IUD
- ▶ At 75 weeks: $\hat{S}(75) = 0.559$
 $\approx 56\%$ still using it
- ▶ At 93 weeks: $\hat{S}(93) = 0.466$
First time the curve drops below 0.5
 $\Rightarrow$ **median** = 93 weeks
- ▶ The curve ends at $t = 107$
(largest observation is censored)



The Nelson-Aalen Estimate — An Alternative Approach

Different formula, same goal

Instead of multiplying $(1 - d_j/n_j)$ factors, we *exponentiate* $(-d_j/n_j)$:

$$\tilde{S}(t) = \prod_{j: t_{(j)} \leq t} e^{-d_j/n_j}$$

Why are the two estimates so similar?

For small x : $e^{-x} \approx 1 - x$

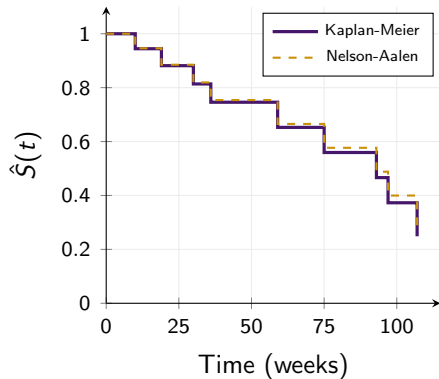
So $e^{-d_j/n_j} \approx 1 - d_j/n_j = (n_j - d_j)/n_j$

The two estimates are nearly identical when d_j is small relative to n_j .

- ▶ $\tilde{S}(t) \geq \hat{S}(t)$ always (since $e^{-x} \geq 1 - x$)
- ▶ Nelson-Aalen is slightly better in very small samples; differences are otherwise negligible
- ▶ **KM remains the standard** for its direct intuitive interpretation

KM vs Nelson-Aalen — IUD Data Comparison

t	$\hat{S}(t)$ (KM)	$\tilde{S}(t)$ (NA)
10	0.9444	0.9460
19	0.8815	0.8850
30	0.8137	0.8194
36	0.7459	0.7539
59	0.6526	0.6653
75	0.5594	0.5768
93	0.4662	0.4882
97	0.3729	0.3997
107	0.2486	0.2864



Max difference: 0.038 — negligible.

Questions?

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